# Research statement

**Sophie Lund Wagner**

*Introduction.*

**My interest in theoretical physics has been driven by a desire to understand the fundamental structure of nature**, which first led me to investigate large-scales within cosmology. Later in my studies, particle physics caught my interest and has naturally pulled me towards quantum field theory. Indeed, I am particularly fascinated by the role of symmetries in shaping physical laws, including how phenomena such as spontaneous symmetry breaking and anomalies play a central role in determining particle behaviour. I wish to deepen my understanding of these principles and ultimately piece together the clues to answer the question **"What symmetries govern physics beyond the Standard Model?"**

I have specifically chosen to apply to the **High Energy Physics Institute in Barcelona (IFAE)** because of its unique position at the interface of theoretical rigor and experimental reality. In what follows, I will first outline my research experience and then present my research plans, which mirror the interdisciplinary excellence of the IFAE-ATLAS group.

*Research experience.*

**My research experience spans fundamental particle physics, cosmology, and computational methods.** Currently, **I am working on my master's thesis**, which focuses on gaining a deeper understanding of fundamental interactions and **beyond-the-Standard-Model physics.** In my research, I explore grand unified theories (GUTs). Rather than proposing a specific gauge group, I study a general landscape of high-energy GUTs and identify preferred models by minimizing their free energy. My project is a continuation of previous work by my supervisor Francesco Sannino [arXiv: 2507.06368], which lacks models including scalars. **Thus, the central goal of my thesis is to extend the landscape approach to GUTs to include scalars,** by determining how different representations of the scalars affect the viability of a candidate GUT. I began by studying chiral gauge theories, such as the Georgi–Glashow [PhysRevLett.32.438] and Bars–Yankielowicz [DOI: 10.1016/0370-2693(81)90664-X] models, introducing scalars in various representations and identifying which combinations can yield completely asymptotically free theories [arXiv:1412.2769, 1605.04712, 1610.03130]; this work is currently being prepared for publication. After this I will compute the free energy for theories with chiral fermions and scalars, with the goal of minimizing it and adding the results to an atlas of possible GUTs that include scalar fields.
In parallel, I am involved in a project on infinite heat order, concerning **high-temperature symmetry non-restoration**, an idea first considered by Weinberg [DOI:10.1103/PhysRevD.9.3357]. This original approach considered non-UV-complete

theories, whereas our work focuses on UV-complete models with this characteristic, building on previous work by my supervisor [arXiv:2012.08428v2]. Through these projects in my master's thesis, **I have developed a strong foundation in quantum field theory.**

Besides my master's thesis, **I have completed two other projects during my master's degree. The first project was on topological anomalies in quantum field theory**, which I completed while on exchange in Canada. The project deepened my understanding of quantum field theory by studying the perturbative chiral anomaly and the non-perturbative Witten anomaly. The Fujikawa and triangle diagrams methods were used, and the chiral anomaly was considered topologically via the Atiyah-Singer index theorem. The Witten anomaly especially opened my eyes to the interplay of topology and the structure of fundamental interaction. **The second project was on percolation models as a comparison to SIR (susceptible, infected, recovered) models initiated at the beginning of the COVID-19 pandemic.** My project was focused on treating infection rate as a phase transition and enhanced my programming skills and model building abilities.

During my bachelor's degree, **I worked on density profiles of dark matter, both theoretically and using machine learning techniques.** This interest culminated in a bachelor's thesis in cosmology, where **I examined whether inhomogeneities in the structure of the universe could explain its accelerated expansion.** Throughout this project, I gained experience with deriving and manipulating complex dynamical equations, implementing numerical models, and analysing how local interactions can produce global effects.

These experiences in analysing gauge theories, exploring topological anomalies, and implementing numerical models have given me a solid foundation in quantum field theory, effective field theories, and computational methods. I am confident that the skills I developed, from deriving analytic constraints to handling complex datasets, will enable me to tackle the research questions on composite dynamics, EFT interpretations, and anomaly constraints that I aim to pursue during my PhD.

*Research plans.*

During my PhD, I want to contribute to research in particle physics. **I yearn to use the framework of quantum field theory to study the interplay of flavour, Higgs dynamics, and beyond-Standard-Model physics.** Ultimately, I aim to clarify the fundamental principles that determine the structure of our Universe. **The INPhINIT fellowship is crucial to this ambition, as it provides the independence and travel budget necessary to foster collaborations with the wider international community.** My specific research plans align closely with the work of Alex Pomarol and Andrea Wulzer, especially in the following three points.

- **Probing Composite Dynamics and Fundamental Constraints.** Building on my master's work on chiral gauge theories and scalars, I aim to investigate composite dynamics and symmetries beyond the Standard Model. Considering composite Higgs models [arXiv:1506.01961], I want to explore how they can be constrained by theoretical consistency conditions. Inspired by Alex Pomarol's recent application of bootstrap techniques to Large-$N_c$ QCD [arXiv:2211.12488], I aim to combine EFT methods with precision Higgs measurements, and to connect fundamental principles of unitarity to phenomenologically testable predictions.
- **Precision EFTs and Future Colliders.** I am deeply motivated by Andrea Wulzer's recent studies on quantifying EFT uncertainties in LHC searches [arXiv:2507.15954]. I want to explore high-energy physics by investigating flavour physics at high-energy muon colliders [arXiv:2509.08132]. I am interested in studying how a multi-TeV muon collider could probe flavour structures and extended-Higgs sectors currently inaccessible at the LHC. This work aligns directly with the group's effort to maximize the discovery potential of current and future experiments.
- **Bootstrapping Anomalies and Axion Physics.** Building on my master's research on topological anomalies, I am eager to investigate how these features manifest in the analytic structure of scattering amplitudes. Inspired by the recent work on "Bootstrapping the chiral-gravitational anomaly" [arXiv:2411.14422], I aim to extend this approach to constrain Axion-Like Particles (ALPs) [arXiv:2003.01100]. My goal is to determine whether these consistency conditions can inform experimental searches, such as the ATLAS experiment and guide future axion measurements.

*Future Aspirations and Impact.*

This fellowship represents a pivotal step toward my long-term goal of becoming an independent academic researcher at the forefront of BSM phenomenology. Beyond research, I am eager to utilize the INPhINIT training program to enhance my leadership and communication skills, ensuring that I can effectively my findings.

I am excited by the prospect of pursuing these research directions at IFAE-Barcelona, where I can combine my background in QFT, symmetries, and computational methods to contribute to advancing our understanding of physics beyond the Standard Model.